

GRD theory package (draft for supervisor review): validity and consistency of the admissibility verdict

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Purpose

Revised 6 October: Lemma C and Corollary 2 close the pooled-null gap; the checks table adds the repository implementation.

The JMLR novelty check found GRD’s empirical side above the journal’s norm and its theory below every comparable CRL paper. The manuscript’s formal results (Propositions 1 and 2, Lemmas 1 and 2, the attribution proposition) are correct but elementary. This note drafts the two results that would change that assessment.

- **Theorem 3 (consistency of the verdict).** Under the paper’s own model, the gate’s certified set equals the set of intervened environments with probability tending to one, so d_{rec} is a consistent estimator of the number of identifiable latent directions, and each certified direction is recovered at rate $\sqrt{d/n}$. The resolution floor of Proposition 1 appears as a necessary condition inside the theorem, not as a separate remark.
- **Theorem 2 (finite-sample validity).** With a pooled-permutation null, every gate p -value is exactly valid, and Benjamini–Yekutieli control of the false discovery rate holds under the arbitrary dependence created by the shared control sample.

Section 5 reports numerical checks of every quantitative claim. Section 6 lists what is not yet proved and what each result costs the paper.

1 Setting and assumptions

Environments $e \in \{0, 1, \dots, m\}$ produce i.i.d. rows $Y_1^{(e)}, \dots, Y_{n_e}^{(e)} \in \mathbb{R}^d$, independent across environments, with covariance Σ_e and precision $\Omega_e = \Sigma_e^{-1}$; $e = 0$ is the control sample. Write $\Delta_e = \Omega_e - \Omega_0$ and $\hat{\Omega}_e$ for the inverse sample covariance. The gate statistic is $T_e = \lambda_{\max}(\hat{\Omega}_e - \hat{\Omega}_0)$, its population value is $\tau_e = \lambda_{\max}(\Delta_e)$, and the p -value is $p_e = (1 + \#\{b : T_e^{(b)} \geq T_e\}) / (B_{\text{boot}} + 1)$ over B_{boot} null draws.

(A1) *Gaussian rows.* Each $Y_i^{(e)} \sim N(\mu_e, \Sigma_e)$.

(A2) *Bounded spectra.* $K^{-1} \leq \lambda_{\min}(\Sigma_e) \leq \lambda_{\max}(\Sigma_e) \leq K$ for all e .

(A3) *Dichotomy.* Every environment is either null, $e \in \mathcal{H}_0$ with $\Sigma_e = \Sigma_0$, or shifted, $e \in \mathcal{H}_1$ with $\tau_e > 0$. Write $k = |\mathcal{H}_1|$, $m_0 = |\mathcal{H}_0|$, and $\gamma = \min_{e \in \mathcal{H}_1} \tau_e$.

(A4) *Sizes.* $n_0 \geq 2 \max_e n_e$, so the disjoint null of the paper applies to every environment.

By Lemma 4 of the paper, $\tau_e > 0$ if and only if $\Sigma_e \not\preceq \Sigma_0$. Environments with $\Sigma_e \succeq \Sigma_0$, $\Sigma_e \neq \Sigma_0$ (pure variance inflation) are excluded by (A3); Remark 2 returns to them.

2 Two lemmas

Lemma A (Precision concentration). *Under (A1) and (A2) there is an absolute constant c such that, for $t > 0$ and $n \geq cK^2(d+t)$, the sample precision of n rows satisfies*

$$\|\widehat{\Omega} - \Omega\|_{\text{op}} \leq \varepsilon(n, t) := cK^3 \sqrt{(d+t)/n}$$

with probability at least $1 - 2e^{-t}$.

Proof. By the Gaussian covariance bound [7, Thm. 5.39, Rem. 5.40], $\|\widehat{\Sigma} - \Sigma\| \leq c_1 K \sqrt{(d+t)/n}$ with probability at least $1 - 2e^{-t}$ (estimating the mean costs one degree of freedom and changes only c_1). If $\|\widehat{\Sigma} - \Sigma\| \leq 1/(2K)$, which the condition on n ensures, then $\lambda_{\min}(\widehat{\Sigma}) \geq 1/(2K)$ and

$$\|\widehat{\Sigma}^{-1} - \Sigma^{-1}\| = \|\widehat{\Sigma}^{-1}(\Sigma - \widehat{\Sigma})\Sigma^{-1}\| \leq 2K \cdot K \cdot \|\widehat{\Sigma} - \Sigma\|.$$

□

Lemma B (Null draws vanish). *Let a null draw split the control sample, uniformly at random and independently of the data, into disjoint parts of sizes n_e and $n_0 - n_e$, and let $T_e^{(b)}$ be the gate statistic of the two parts. Under (A1), (A2), and (A4), each draw satisfies $P(T_e^{(b)} > 2\varepsilon(n_e, t)) \leq 4e^{-t}$. For the $(1 - \alpha)$ quantile $\widehat{c}_e(\alpha)$ of B_{boot} draws, $P(\widehat{c}_e(\alpha) > 2\varepsilon(n_e, t)) \leq 4e^{-t}/\alpha$.*

Proof. Because the split is independent of the data and the control rows are i.i.d., the two parts are, unconditionally, independent i.i.d. samples from $N(\mu_0, \Sigma_0)$ of sizes n_e and $n_0 - n_e \geq n_e$. Weyl's inequality gives $T_e^{(b)} \leq \|\widehat{\Omega}_A - \Omega_0\| + \|\widehat{\Omega}_B - \Omega_0\|$, and Lemma A bounds each term. For the quantile, the fraction F of draws exceeding 2ε has $EF \leq 4e^{-t}$, and $\widehat{c}_e(\alpha) > 2\varepsilon$ requires $F > \alpha$; apply Markov's inequality. □

3 Theorem 3: consistency of the admissibility verdict

Theorem 3 (Verdict consistency). *Assume (A1) to (A4), let $\widehat{C}$ be the BH-certified set at level q , and fix $\delta \in (0, 1)$. Set $t = \log\{4(m + kB_{\text{boot}})/\delta\}$ and $\bar{\varepsilon} = \max_e \varepsilon(n_e, t)$, and assume $\min_e n_e \geq cK^2(d+t)$ as in Lemma A.*

(i) **Accurate statistics.** *With probability at least $1 - \delta$, $|T_e - \tau_e| \leq 2\bar{\varepsilon}$ for every e , and no null draw of any shifted environment exceeds $2\bar{\varepsilon}$.*

(ii) **Sure screening.** *If $\gamma > 4\bar{\varepsilon}$ and*

$$(B_{\text{boot}} + 1)kq \geq m, \tag{1}$$

then $P(\mathcal{H}_1 \subseteq \widehat{C}) \geq 1 - \delta$. Conversely, if $(B_{\text{boot}} + 1)kq < m$ and every null p -value exceeds q , then $\widehat{C} = \emptyset$ however large γ is, so condition (1) cannot be weakened.

(iii) **Exact recovery.** If, in addition, the null p -values are valid, $P(p_e \leq u) \leq u$ for $e \in \mathcal{H}_0$ and $u \in [0, 1]$, then

$$P(\widehat{C} = \mathcal{H}_1) \geq 1 - \delta - m_0 q.$$

Hence, if $q = q_n \rightarrow 0$, $B_{\text{boot}} \geq m/(kq_n)$, and $\gamma/\bar{\varepsilon} \rightarrow \infty$, the verdict $d_{\text{rec}} = \min\{|\widehat{C}|, d_\star\}$ converges in probability to $\min\{k, d_\star\}$.

Proof. (i) Apply Lemma A to each of the $m + 1$ samples with probability at least $1 - 2e^{-t}$ each and Weyl's inequality, $|T_e - \tau_e| \leq \|\widehat{\Omega}_e - \Omega_e\| + \|\widehat{\Omega}_0 - \Omega_0\|$; apply Lemma B to the kB_{boot} draws of shifted environments. A union bound over at most $2(m + 1) + 4kB_{\text{boot}} \leq 4(m + kB_{\text{boot}})$ failure events of probability e^{-t} gives $1 - \delta$.

(ii) On the event of (i), every $e \in \mathcal{H}_1$ has $T_e \geq \gamma - 2\bar{\varepsilon} > 2\bar{\varepsilon} \geq T_e^{(b)}$ for every draw b , so $p_e = 1/(B_{\text{boot}} + 1)$. The k smallest p -values are then all equal to $1/(B_{\text{boot}} + 1)$, and by (1) $p_{(k)} = 1/(B_{\text{boot}} + 1) \leq kq/m$. The BH rejection count R is the largest j with $p_{(j)} \leq jq/m$, so $R \geq k$, and BH rejects every p -value at most $p_{(R)} \geq 1/(B_{\text{boot}} + 1)$, which includes all of $\mathcal{H}_1$. For the converse, by Proposition 1 of the paper any nonempty BH set has at least $m/\{q(B_{\text{boot}} + 1)\} > k$ elements, so it must contain a null environment; but a rejected p -value is at most $Rq/m \leq q$, and every null p -value exceeds q . Hence $R = 0$.

(iii) BH rejects e only if $p_e \leq Rq/m \leq q$. For $e \in \mathcal{H}_0$, validity gives $P(p_e \leq q) \leq q$, so $P(\widehat{C} \cap \mathcal{H}_0 \neq \emptyset) \leq m_0 q$. Combine with (ii). The limit statement follows because δ can be taken to zero with $t = \log\{4(m + kB_{\text{boot}})/\delta\}$ growing only logarithmically while $\gamma/\bar{\varepsilon} \rightarrow \infty$. $\square$

Parts (i) and (ii) hold for the paper's disjoint null as it is. Part (iii) needs valid null p -values, which the disjoint null provides only approximately; Theorem 2 supplies them for the centred pooled-permutation null, and Lemma C shows that parts (i) and (ii) also hold for that null, with $\bar{\varepsilon}$ replaced by a bound larger by a logarithmic factor.

Remark 1 (What the theorem adds to Proposition 1). Proposition 1 shows that a coarse null forces abstention. Theorem 3(ii) shows that the same inequality, $(B_{\text{boot}} + 1)kq \geq m$, is also sufficient: once the statistic separates, the resolution of the null is the only remaining obstacle to certifying every shifted environment. The Norman screen is an instance: with $m = 101$, $k = 1$, and $q = 0.05$, condition (1) needs $B_{\text{boot}} + 1 \geq 2,020$; it fails at 500 draws and holds at 9,999.

Corollary 1 (Link to identifiability). *Take the model of Section 2.1 of the paper, $Y^{(e)} = RZ^{(e)} + \eta^{(e)}$ with a shared invertible R , a shared noise covariance $\text{Cov}(\eta^{(e)}) = \sigma^2 I$, and environments that are either unintervened or perform one atomic hard intervention on a target $i(e)$, with $\lambda'_i < \lambda_i$ whenever the target has no parents. Then:*

(a) $\mathcal{H}_1$ is exactly the set of intervened environments, so under the conditions of Theorem 3(iii) $d_{\text{rec}} \rightarrow \min\{\#\text{intervened targets}, d_\star\}$ in probability, the number of latent directions that the recovery theorem identifies, capped at the working dimension.

(b) For every certified e , with probability at least $1 - \delta$, the recovered direction satisfies

$$\sin \angle(v_{\max}(\widehat{\Omega}_e - \widehat{\Omega}_0), v_{\max}(\Delta_e)) \leq 4\bar{\varepsilon}/\gamma,$$

and, when $\sigma = 0$, $v_{\max}(\Delta_e)$ lies within the angle of Equation (8) of the paper from the target's unmixing row, and equals it when the target has no parents.

Proof. (a) The latent precision changes by $\Theta_e - \Theta_0 = \lambda'_i{}^{-1}e_i e_i^\top - \lambda_i{}^{-1}(e_i - b_i)(e_i - b_i)^\top$ (Proposition 2 of the paper). If $b_i \neq 0$, take $v \perp e_i - b_i$ with $v^\top e_i \neq 0$; then $v^\top(\Theta_e - \Theta_0)v = \lambda'_i{}^{-1}(v^\top e_i)^2 > 0$.

If $b_i = 0$, $\Theta_e - \Theta_0 = (\lambda_i'^{-1} - \lambda_i^{-1})e_i e_i^\top$, which has a positive eigenvalue when $\lambda_i' < \lambda_i$. Either way $\Sigma_{Z,e} \not\preceq \Sigma_{Z,0}$ by Lemma 4 of the paper. Congruence by R and adding the common $\sigma^2 I$ preserve the Loewner relation of the two covariances, because $\Sigma_e - \Sigma_0 = R(\Sigma_{Z,e} - \Sigma_{Z,0})R^\top$; so $\tau_e > 0$. Unintervened environments have $\Sigma_e = \Sigma_0$. (b) For positive definite A, B , $A^{-1} - B^{-1} = A^{-1/2}(I - A^{1/2}B^{-1}A^{1/2})A^{-1/2}$ and $B - A = A^{1/2}(A^{-1/2}BA^{-1/2} - I)A^{1/2}$, so $A^{-1} - B^{-1}$ and $B - A$ have the same inertia. Applying this twice, Δ_e has the inertia of $\Sigma_0 - \Sigma_e = R(\Sigma_{Z,0} - \Sigma_{Z,e})R^\top$, hence of $\Theta_e - \Theta_0$: rank at most two and at most one positive eigenvalue, with or without observation noise. Its eigengap satisfies $\lambda_1 - \lambda_2 \geq \tau_e \geq \gamma$, and the Davis–Kahan variant of Yu et al. [8] gives $\sin \theta \leq 2\|E\|/\gamma$ with $\|E\| \leq 2\bar{\varepsilon}$ on the event of Theorem 3(i). The noiseless statement is Proposition 2 of the paper. $\square$

Remark 2 (Off the model). Proposition 5 of the paper shows that measurement changes can also have $\tau_e > 0$. Theorem 3 does not distinguish them: off the model, d_{rec} is a consistent estimator of the number of *detectable* directions, and equals the number of identifiable directions only under the corollary’s model. Environments with pure variance inflation, $\Sigma_e \succeq \Sigma_0$, have $\tau_e \leq 0$ and are not certified with probability tending to one; they are exactly the invisible measurement changes of Proposition 5(i).

4 Theorem 2: exact validity with a pooled-permutation null

The paper’s disjoint null draws both pseudo-samples from controls. The pooled-permutation null instead pools $Y^{(e)}$ and $Y^{(0)}$, relabels the $n_e + n_0$ rows uniformly at random into groups of sizes n_e and n_0 , and recomputes the statistic. Because Perturb-seq perturbations often shift means, each group is first centred at its own mean; Section 5 shows why this matters.

Theorem 2 (Exact validity and dependence-robust FDR). *Suppose that for every $e \in \mathcal{H}_0$ the rows of $Y^{(e)}$ and $Y^{(0)}$ are jointly exchangeable (for example, i.i.d. from one distribution). With pooled-permutation p -values:*

- (i) $P(p_e \leq u) \leq u$ for every $e \in \mathcal{H}_0$ and $u \in [0, 1]$, for any $B_{\text{boot}} \geq 1$ and any sample sizes;
- (ii) the Benjamini–Yekutieli procedure at level q satisfies $\text{FDR} \leq q m_0/m$, whatever the dependence among the p -values, including the dependence induced by the shared control sample.

Proof. (i) Under exchangeability, the observed labelling is one uniformly random element of the relabelling group, independent of the B_{boot} random relabellings, so the observed statistic is exchangeable with the B_{boot} draws; the p -value $(1 + \#)/(B_{\text{boot}} + 1)$ is then super-uniform [5, 3]. (ii) Benjamini–Yekutieli control holds for any joint law of valid p -values [1, Theorem 1.3]. $\square$

Remark 3 (Centring). Exchangeability, and so exactness, holds for environments with the same distribution as the controls. An environment that shifts only its mean has $\tau_e = 0$ but is not exchangeable with the controls, and without centring the pooled test rejects it: in our check, 10% of such environments had $p \leq 0.05$. Centring each group at its own mean removes the shift; the centred residuals are exchangeable only approximately (each group’s residuals sum to zero), so validity becomes approximate, with error of order $1/n_e$. The same check gives rejection rates of 3.3% to 7.3% at the 5% level after centring (Section 5).

The pooled null keeps the consistency of Theorem 3. The difference from Lemma B is that under $\mathcal{H}_1$ the pool mixes two distributions, so its subsets are not samples from either; the point is that both pseudo-groups are random subsets of the same pool, and therefore estimate the same matrix.

Lemma C (Pooled null draws vanish, for every environment). *Assume (A1), (A2), and (A4), and let $N = n_e + n_0$. There are absolute constants c, c' such that, for $\eta \in (0, 1)$ and $n_e \geq cK^4(d + \log(N/\eta)) \log(2d/\eta)$, a centred pooled-permutation draw satisfies*

$$P\left(T_e^{(b)} > \varepsilon'(n_e, \eta)\right) \leq c'\eta, \quad \varepsilon'(n_e, \eta) := cK^3 \sqrt{\frac{\{d + \log(N/\eta)\} \log(2d/\eta)}{n_e}},$$

whether $e \in \mathcal{H}_0$ or $e \in \mathcal{H}_1$.

Proof. Condition on the data and write $z_1, \dots, z_N$ for the centred pooled rows, $S_P = N^{-1} \sum_i z_i z_i^\top$, and $L = \max_i \|z_i\|^2$. A draw splits the pool uniformly into A (n_e rows) and B (n_0 rows); let S_A, S_B be the corresponding averages of $z_i z_i^\top$. Since $n_e S_A + n_0 S_B = N S_P$, we have $S_B - S_P = -(n_e/n_0)(S_A - S_P)$, so it suffices to control S_A .

Step 1: the pool. By Lemma A applied to each group and $n_0 \geq 2n_e$, with probability at least $1 - 4\eta$, $\|S_P\| \leq 2K$ and, since $\lambda_{\min}(\widehat{\Sigma}_0) \geq 1/(2K)$ on the event of Lemma A, $\lambda_{\min}(S_P) \geq (n_0/N)\lambda_{\min}(\widehat{\Sigma}_0) \geq 1/(3K)$. By a χ^2 tail bound [4] and a union bound over the N rows, $L \leq cK\{d + \log(N/\eta)\}$ with probability at least $1 - 2\eta$.

Step 2: one draw. S_A is the average of n_e matrices sampled without replacement from $\{z_i z_i^\top\}$, whose mean is S_P . Sampling without replacement satisfies the same matrix moment-generating-function bounds as sampling with replacement [2], so matrix Bernstein [6], with summands bounded by $2L$ and variance proxy $L\|S_P\|$, gives $\|S_A - S_P\| \leq \sqrt{2L\|S_P\| \log(2d/\eta)/n_e} + (4L/3) \log(2d/\eta)/n_e$ with probability at least $1 - \eta$. The same argument on the Hermitian dilation of the subset mean m_A (the pool has mean zero) gives $\|m_A\|^2 \leq cL \log(2d/\eta)/n_e$. The sample covariance of A is $\frac{n_e}{n_e-1}(S_A - m_A m_A^\top)$, so on these events it lies within $\rho = cK \sqrt{\{d + \log(N/\eta)\} \log(2d/\eta)/n_e}$ of S_P , and the sample covariance of B lies within $(n_e/n_0)\rho + O(\rho^2) \leq \rho$ of it.

Step 3: inversion. The condition on n_e makes $\rho \leq \lambda_{\min}(S_P)/2$, so both inverses lie within $2\{3K\}^2 \rho$ of S_P^{-1} , and by Weyl's inequality $T_e^{(b)} \leq \|\widehat{\Omega}_A - \widehat{\Omega}_B\| \leq 36K^2 \rho = \varepsilon'$. No step used $\Sigma_e = \Sigma_0$: only the spectral bounds (A2) and $n_0 \geq 2n_e$. $\square$

Corollary 2 (Theorem 3 under the pooled null). *With the centred pooled null, parts (i) and (ii) of Theorem 3 hold with $\bar{\varepsilon}$ replaced by $\bar{\varepsilon}' = \max_e \varepsilon'(n_e, \eta)$ for $\eta = \delta/\{c'(m + kB_{\text{boot}})\}$, and part (iii) holds with the null p -values of Theorem 2: exactly valid for environments with the same distribution as the controls, approximately valid after centring for environments that shift only their mean.*

Proof. Replace Lemma B by Lemma C in the proof of Theorem 3; the union bound has the same number of terms. $\square$

5 Numerical checks

Every quantitative claim was checked on the paper's own model ($d = 10$, random sparse DAG, random invertible R , hard interventions to noise variance 0.1 or 0.05), with the repository's gate code (`src/gate/precision_readout.py`) for the statistic, the disjoint null, and BH.

6 Gaps and cost to the paper

1. **Pooled-null consistency.** Closed by Lemma C and Corollary 2 (6 October). The bound is larger than the disjoint one by a factor $\sqrt{\log(2d/\eta)}$ and uses only (A2) and $n_0 \geq 2n_e$.

Claim	Check	Result
Lemma A and Theorem 3(i): $ T_e - \tau_e = O_P(\sqrt{d/n})$	Intervened target with parents, $n_0 = 20,000$, $n_e \in \{250, \dots, 8,000\}$, 20 replicates; log-log slope of the median error.	Slope -0.57 ; median error 2.50 at $n_e = 250$ to 0.35 at 8,000. Consistent with $n^{-1/2}$.
Lemma B: null threshold $= O_P(\sqrt{d/n})$	Same design, 95% quantile of 200 disjoint-null draws.	Slope -0.53 ; quantile 7.89 to 1.29. Consistent with $n^{-1/2}$.
Theorem 3(ii): condition (1) is sharp	$m = 21$, $k = 1$ strong environment, $q = 0.05$, so the threshold is $B_{\text{boot}} + 1 \geq 420$; 20 replicates at $B_{\text{boot}} = 399$ and $B_{\text{boot}} = 499$.	The shifted environment's p -value hit the floor in 20/20 runs at $B_{\text{boot}} = 399$ and 19/20 at 499. It was certified in 1/20 runs at 399 (only alongside a null environment with a near-floor p -value, as the converse allows) and in 19/20 at 499, exactly the runs at the floor; no null certified at 499.
Theorem 2(i): pooled p -values valid with shared controls	$m = 10$ null environments sharing one control sample ($n_e = 300$, $n_0 = 1,500$, $B_{\text{boot}} = 99$), 120 replicates.	$P(p \leq 0.05) = 0.042$, $P(p \leq 0.10) = 0.088$, $P(p \leq 0.20) = 0.18$. Any rejection with all environments null at $q = 0.2$: BH 0.15, BY 0.
Centring remark: mean-only shifts	Pooled null with a Mahalanobis mean shift of 1, $n_e = 300$, $n_0 = 1,500$, 150 replicates; then the centred version with shifts of 0, 1, and 3.	Uncentred: $P(p \leq 0.05) = 0.10$, invalid. Centred: 0.033, 0.073, 0.053 (standard error about 0.018).
Repository implementation (commit 8ebe72d)	<code>experiments/e8_pooled_null.py</code> <code>--validity-check</code> : the same two designs run through the committed gate code, plus an exact-equality regression of the default (disjoint) path against the previous module.	Null: $P(p \leq 0.05) = 0.048$, $P(p \leq 0.20) = 0.213$, any BY rejection 0.008. Mean shift: 0.053. Default path byte-identical.

- Code and reruns.** The centred pooled null and BY are in the gate (commit 8ebe72d; default path unchanged). The A100 rerun of the three Perturb-seq panels at $B_{\text{boot}} = 9,999$ with this null (E8) is running; its BH and BY sets replace the disjoint-null sets as the paper's primary real-data decision.
- Assumption (A3).** The dichotomy excludes environments that shift only slightly ($0 < \tau_e \leq 4\bar{\varepsilon}$). A version with three classes (null, weak, strong) and a guarantee only for the strong class is straightforward and more honest for real data.
- Gaussianity.** (A1) can be relaxed to sub-Gaussian rows with the same rate; heavy-tailed single-cell data are not covered.
- Novelty.** The ingredients (Weyl, covariance concentration, Davis–Kahan, randomization tests, BH and BY) are standard. What is new is the statement: a consistency guarantee for an admissibility verdict that is coupled to an identifiability theorem, with the null resolution as an explicit necessary and sufficient condition. On the rubric used for the JMLR check this moves the theory score from 2 to about 3, not to the 4 or 5 of identifiability papers.

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